Expected Value KEY

Name _____

This table shows the probability of each number of goals a soccer team scores in a game. On average, how many goals does the team score?

x (goals)	0	1	2	3	4
$P(x)$	0.1	0.3	0.3	0.2	0.1
$x \cdot P(x)$	0	0.3	0.6	0.6	0.4

Expected value: $E(X) = \Sigma\, x \cdot P(x)$

$E(X)$ means the expected value of X. Multiply each value by its probability. Then add them all up (Σ means add up). The expected value is also called the **mean** of X, written μ_X.

$E(X) = 0 + 0.3 + 0.6 + 0.6 + 0.4 =$ **1.9** goals

What it means: over many games, the team scores about 1.9 goals per game on average. No single game has 1.9 goals. The expected value is a **long-run average**.

Fair decisions. A method is **fair** if every person has the same probability of being chosen.

Ana, Ben, and Cai need to decide who goes first.

Method	**Fair?**
Roll a number cube. 1 or 2: Ana, 3 or 4: Ben, 5 or 6: Cai.	Yes. Each has $\frac{2}{6} = \frac{1}{3}$.
Flip 2 coins. HH: Ana, TT: Ben, one of each: Cai.	No. Ana $\frac{1}{4}$, Ben $\frac{1}{4}$, Cai $\frac{2}{4} = \frac{1}{2}$.

Fair games. A game is fair if the expected value of the money you win or lose is **0**.

Example: Roll a number cube. You win \$3 if you roll a 6. You pay \$1 for any other number. Let X = the money you win or lose in one game.

$E(X) = 3 \cdot \frac{1}{6} + (-1) \cdot \frac{5}{6} = \frac{3}{6} - \frac{5}{6} = -\frac{2}{6} \approx -\$0.33$

On average you lose about 33 cents a game. The game is **not** fair.

Work with a partner. Compare answers when you finish each problem.

1. Find the expected value.

(a)

x	1	2	3
$P(x)$	0.2	0.5	0.3

$1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$

(b)

x	0	1	2	3
$P(x)$	0.3	0.4	0.2	0.1

$0 + 0.4 + 0.4 + 0.3 = 1.1$

2. Flip a coin 3 times. X = the number of heads has $P(0) = \frac{1}{8}$, $P(1) = \frac{3}{8}$, $P(2) = \frac{3}{8}$, $P(3) = \frac{1}{8}$.

   (a) Find $E(X)$. $0 + \frac{3}{8} + \frac{6}{8} + \frac{3}{8} = \frac{12}{8} = 1.5$

   (b) Explain what $E(X)$ means. If you flip 3 coins many times, you get 1.5 heads per set of 3 on average.

3. The number of pets per household is X, with $P(0) = 0.30$, $P(1) = 0.40$, $P(2) = 0.20$, $P(3) = 0.10$. A student says, "The expected value is 1.1, so a household can have 1.1 pets." What is wrong with this?

   No household has 1.1 pets. 1.1 is the average over many households.

4. Four friends use a spinner with 4 equal parts to pick who chooses the movie. Each friend gets one part. Then they switch to rolling a number cube: 1: Ana, 2: Ben, 3: Cai, 4 to 6: Dee.

   (a) Is the spinner fair? Explain. Yes. Each friend has $\frac{1}{4}$.

   (b) Is the number cube fair? Explain. No. Dee has $\frac{3}{6} = \frac{1}{2}$. The others have $\frac{1}{6}$.

   (c) Change the number cube method to make it fair.
   Example: 1: Ana, 2: Ben, 3: Cai, 4: Dee, and roll again on 5 or 6.

5. A game: roll a number cube. You win \$5 if you roll a 6. You pay \$1 for any other number. Let X = the money you win or lose in one game.

   (a) Find the expected value. $5 \cdot \frac{1}{6} + (-1) \cdot \frac{5}{6} = \frac{5}{6} - \frac{5}{6} = 0$

   (b) Is the game fair? Explain. Yes. The expected value is 0, so no one wins or loses on average.

6. In the notes example, you win \$3 if you roll a 6. How much should you pay for any other number to make the game fair?

   Let c = what you pay. $3 \cdot \frac{1}{6} - c \cdot \frac{5}{6} = 0$, so $5c = 3$ and $c = \$0.60$.